

ISLAMIC UNIVERSITY OF TECHNOLOGY

Board Bazar, Gazipur, Dhaka.



Software Design and Architectures - SWE 4602 **Structural , Behavioral , and UI Design of BookHaven**

Sixth Semester
Department of Computer Science and Engineering
BSc. in Software Engineering

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1 Introduction

In the era of digital transformation, traditional methods of purchasing books are rapidly being replaced by online platforms that offer convenience, efficiency, and broader accessibility. Online bookstores have become an integral part of modern e-commerce, serving a wide range of users from avid readers to academic researchers. This growing demand for digital book shopping necessitates the development of robust, scalable, and user-friendly systems that can efficiently handle the dynamics of inventory, user interactions, and transactions.

This project centers around the comprehensive design of a **Bookstore Management System**, focusing on three core aspects of software design: **Structural**, **Behavioral**, and **User Interface (UI) Design**. The system is designed to cater to two primary user groups: general users (customers) and administrators. General users can register, log in, search and browse books by categories, add items to a shopping cart, and complete purchases through a secure payment process. On the other hand, administrators can manage book inventories, update listings, monitor transactions, and maintain the overall catalog.

To systematically approach the development of this system, we have utilized key principles from software architecture and design patterns. The **structural design** encompasses diagrams such as the **Class Diagram**, which represents the static relationships among entities; the **Object Diagram**, which illustrates real-time instances; the **Component Diagram**, which visualizes modular structure; and the **Deployment Diagram**, which outlines the system's physical deployment across servers and devices.

The **behavioral design** aspects of the system include a **Use Case Diagram** to outline system interactions, **Activity Diagrams** to model workflows, and a **Sequence Diagram** to demonstrate the chronological interaction between system components and users. These diagrams help in visualizing and validating the dynamic behavior of the system under various scenarios.

Furthermore, the **User Interface Design** section involves detailed user and task analysis to ensure the system is intuitive and accessible to a broad audience. It includes mockups and demo screens for all key interfaces such as registration, login, homepage, book browsing, shopping cart, checkout, payment, and admin dashboard. These UI components are designed with a focus on simplicity, consistency, and responsiveness.

This report provides a holistic view of the entire design process, highlighting individual and team contributions. It reflects a collaborative effort to integrate theoretical knowledge with practical application by employing modern design techniques, industry-standard modeling tools, and best practices in software engineering. Ultimately, the goal is to deliver a seamless and efficient digital experience for both customers and administrators while ensuring the system's maintainability, scalability, and security.

2 Team Members And Contribution

2.1 Team Members

1. Namisa Najah Raisa - 210042112
2. Nazifa Tasneem - 210042114
3. Irfan Hakim Bhuiyan - 210042158

2.2 Contribution

Contribution	Student ID
Introduction	210042158
Class Diagram	210042112 and 210042114 and 210042158
Object Diagram	210042112
Component Diagram	210042114
Deployment Diagram	210042114
Use Case Diagram	210042112 and 210042114
Activity Diagram for different systems	210042112
Sequence Diagram	210042158
User Analysis	210042158
Task Analysis	User Registration: 210042158 User Login and Admin's inventory management: 210042114 Book order and Shopping cart: 210042112
Demo Interfaces	Login,signup,home page: 210042158 Browse books, shopping cart, checkout pages: 210042112 Payment,Admin's dashboard, admin's book management pages: 210042114

Table 1: Team Member Contributions

3 Structural Design

3.1 Class Diagram

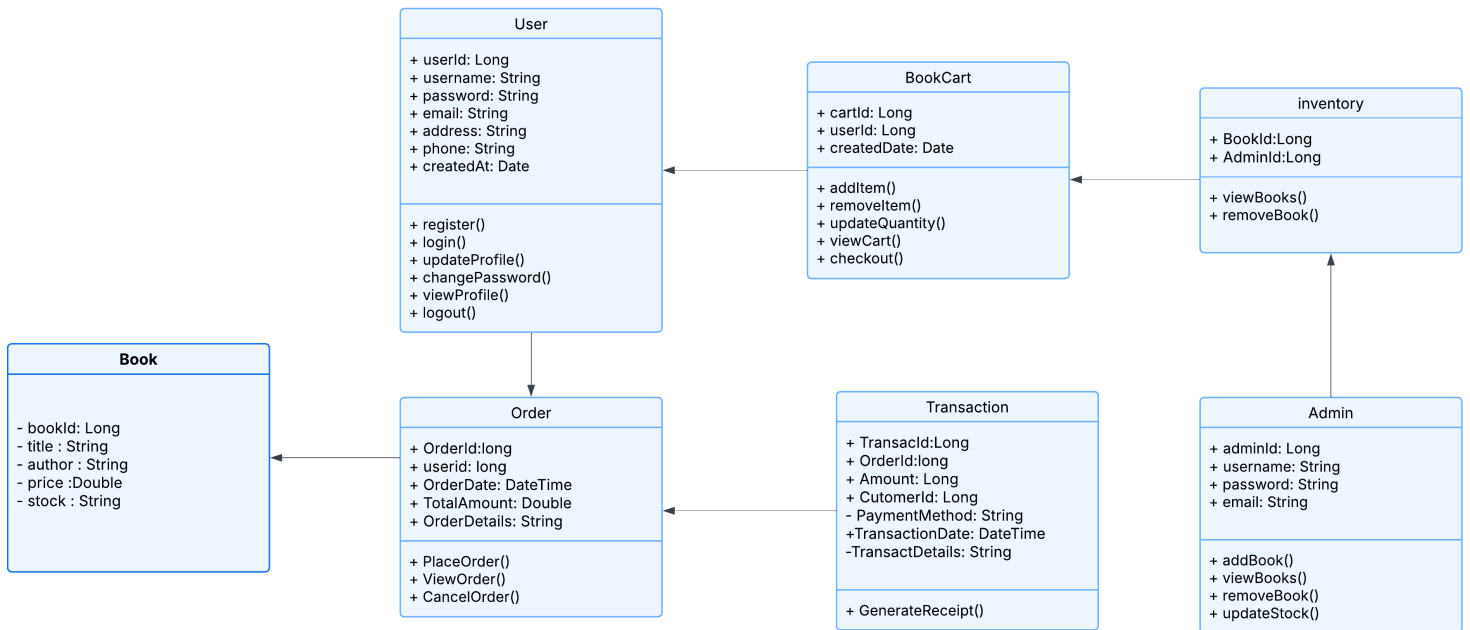


Figure 1: Class Diagram

3.2 Object Diagram

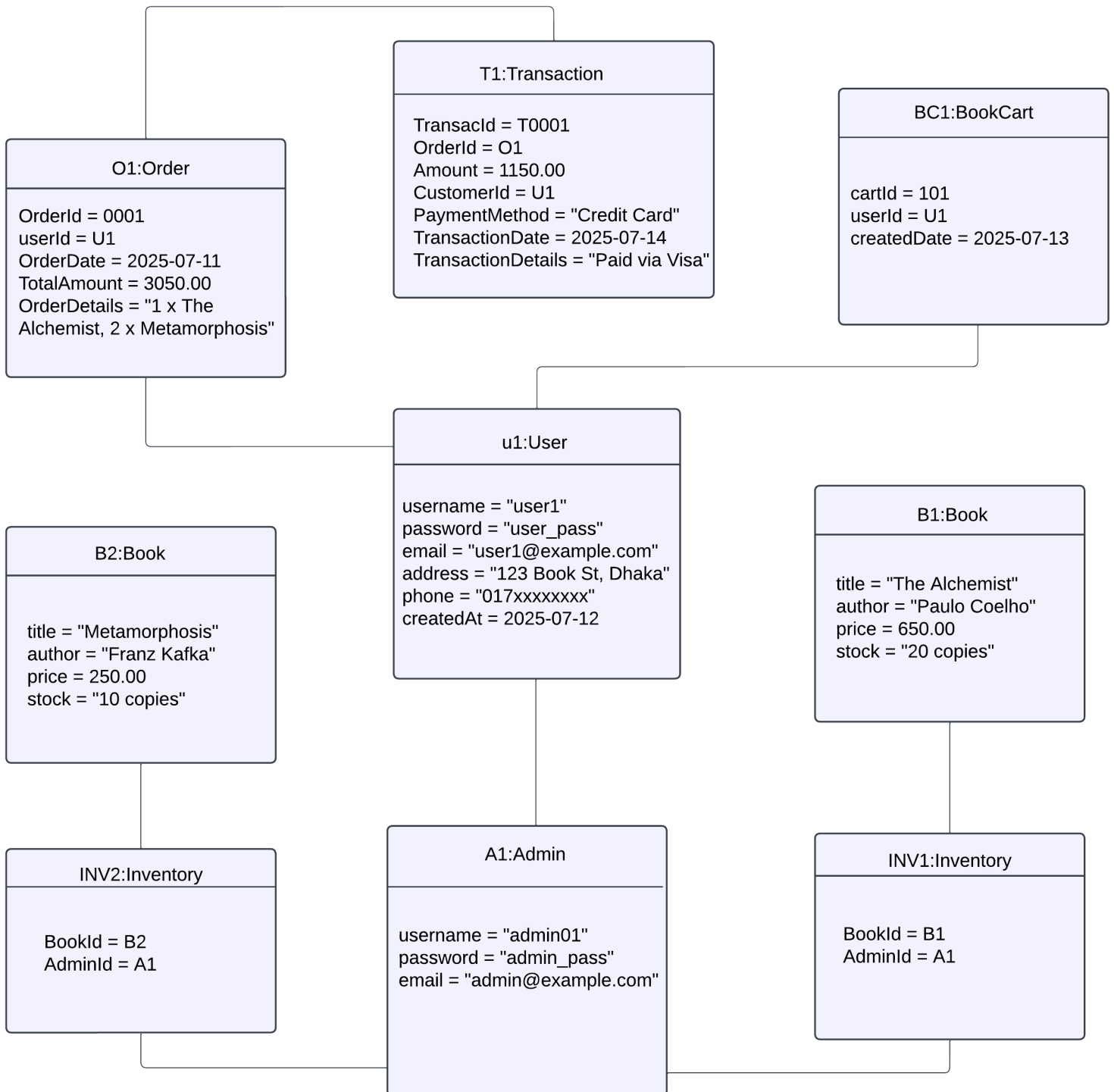


Figure 2: Object Diagram

3.3 Component Diagram

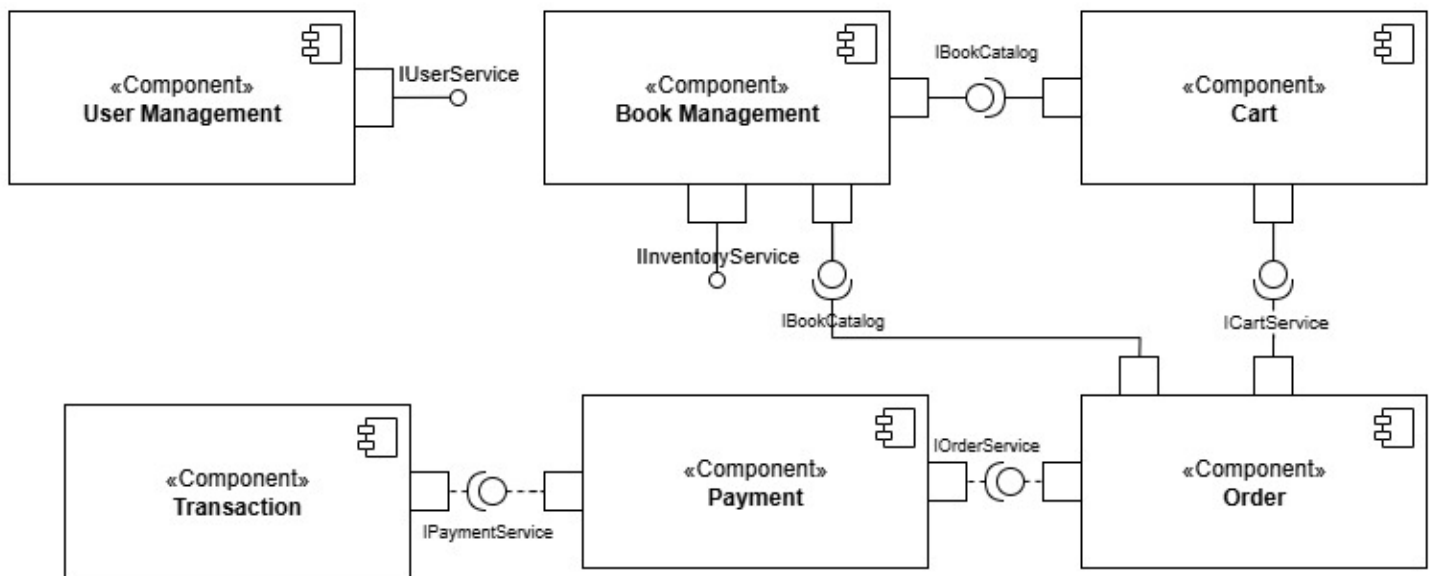


Figure 3: Component Diagram

3.4 Deployment Diagram

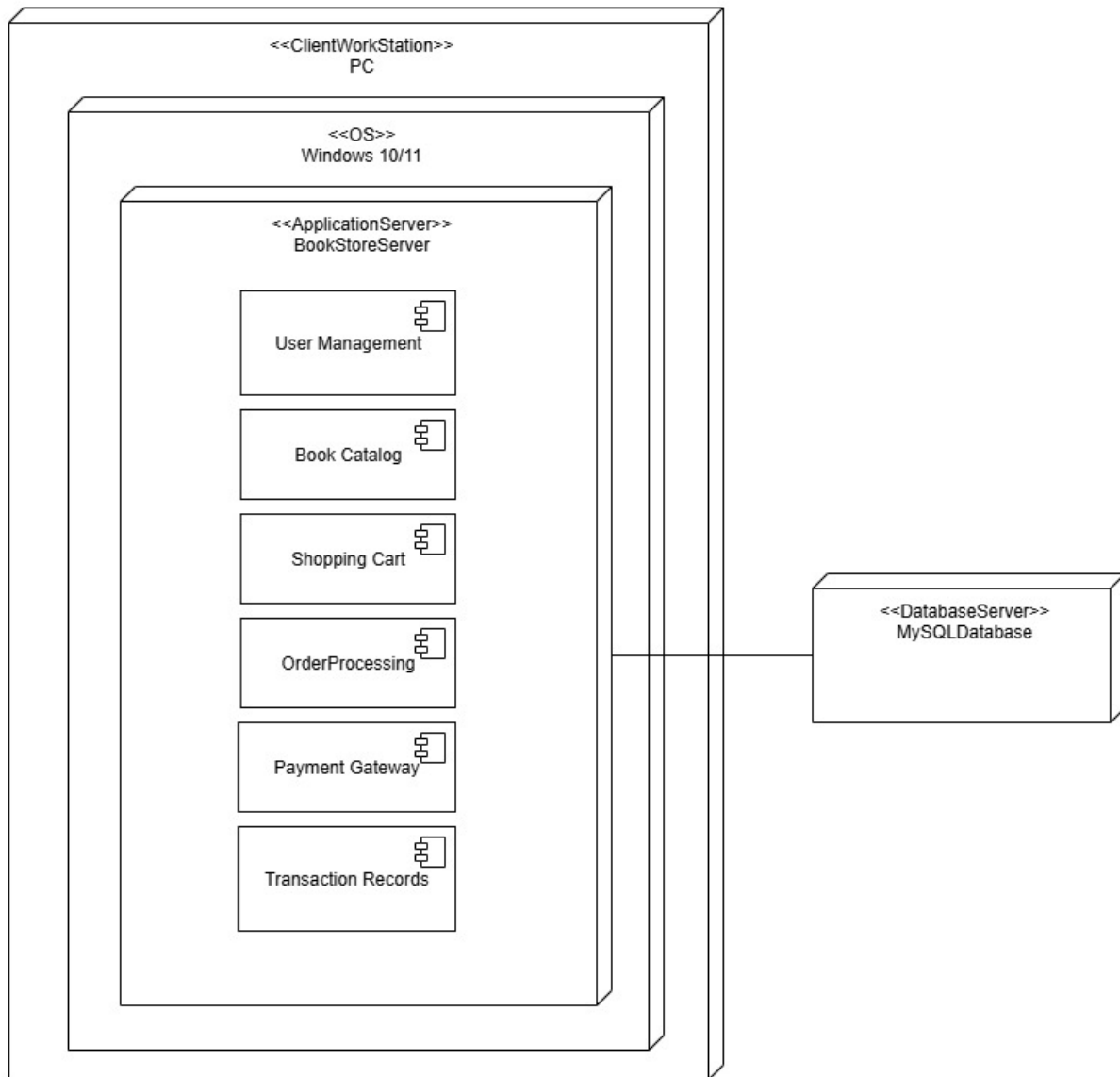


Figure 4: Deployment Diagram

4.2 Activity Diagram

4.2.1 User registration System

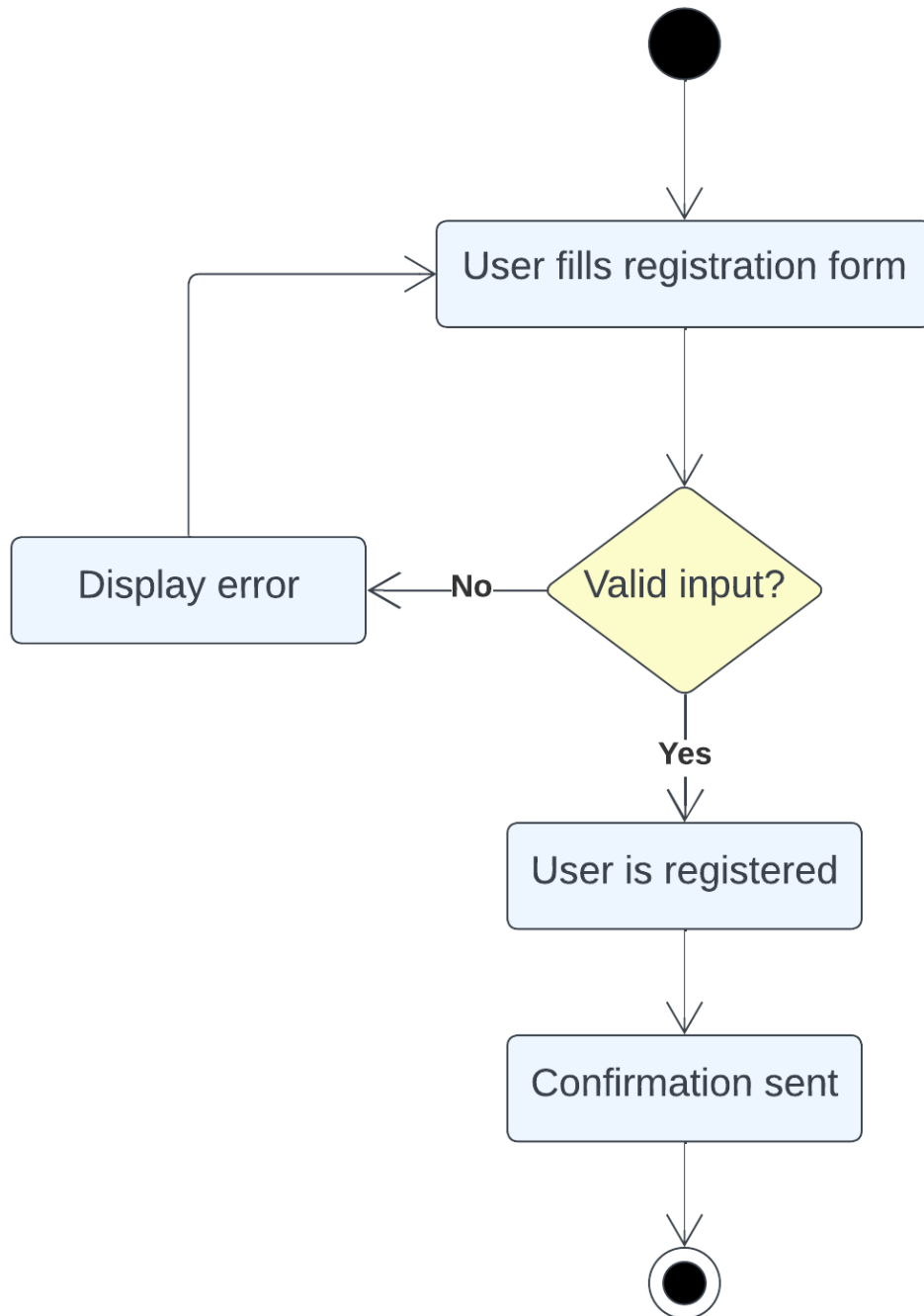


Figure 6: User registration System

4.2.2 User Login System

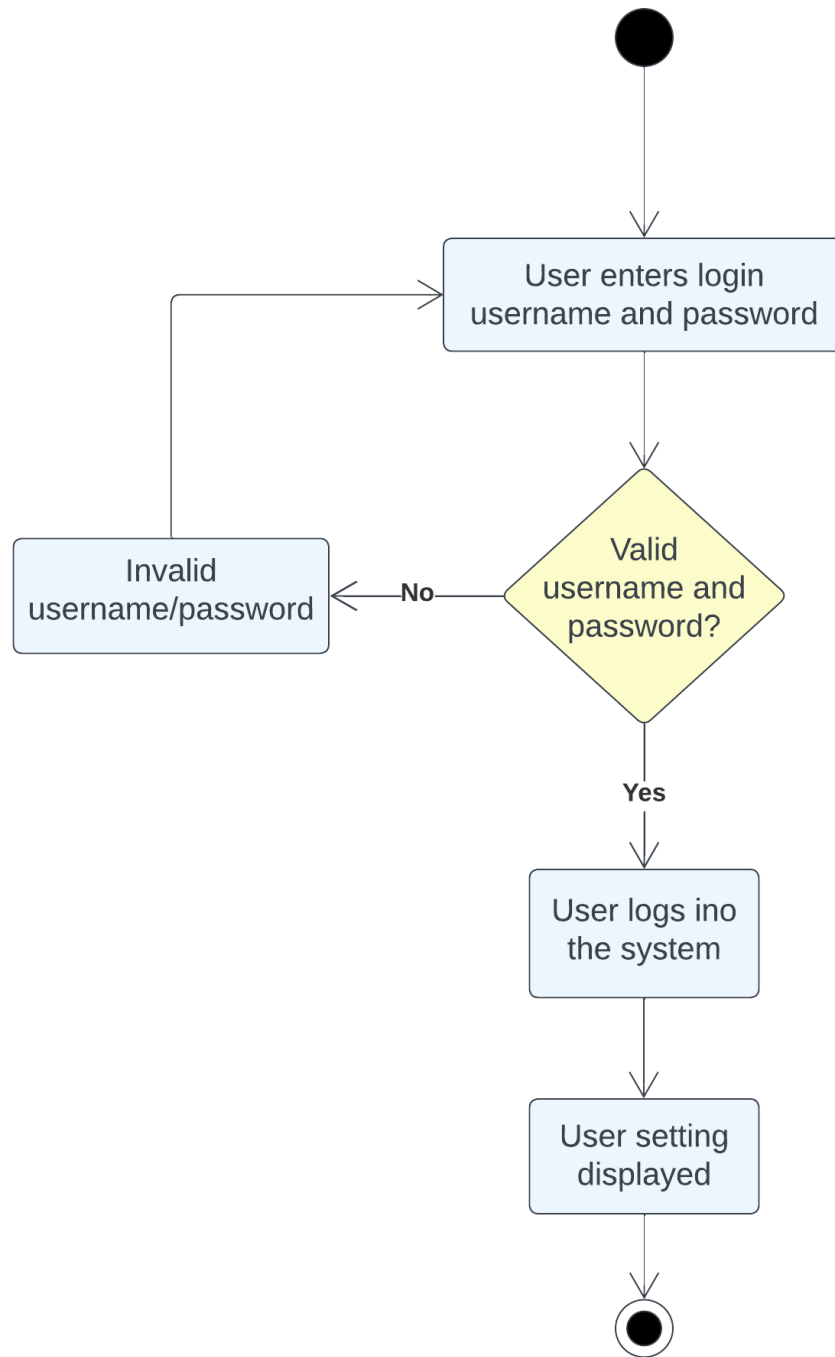


Figure 7: User login System

4.2.3 Admin: Book Management system

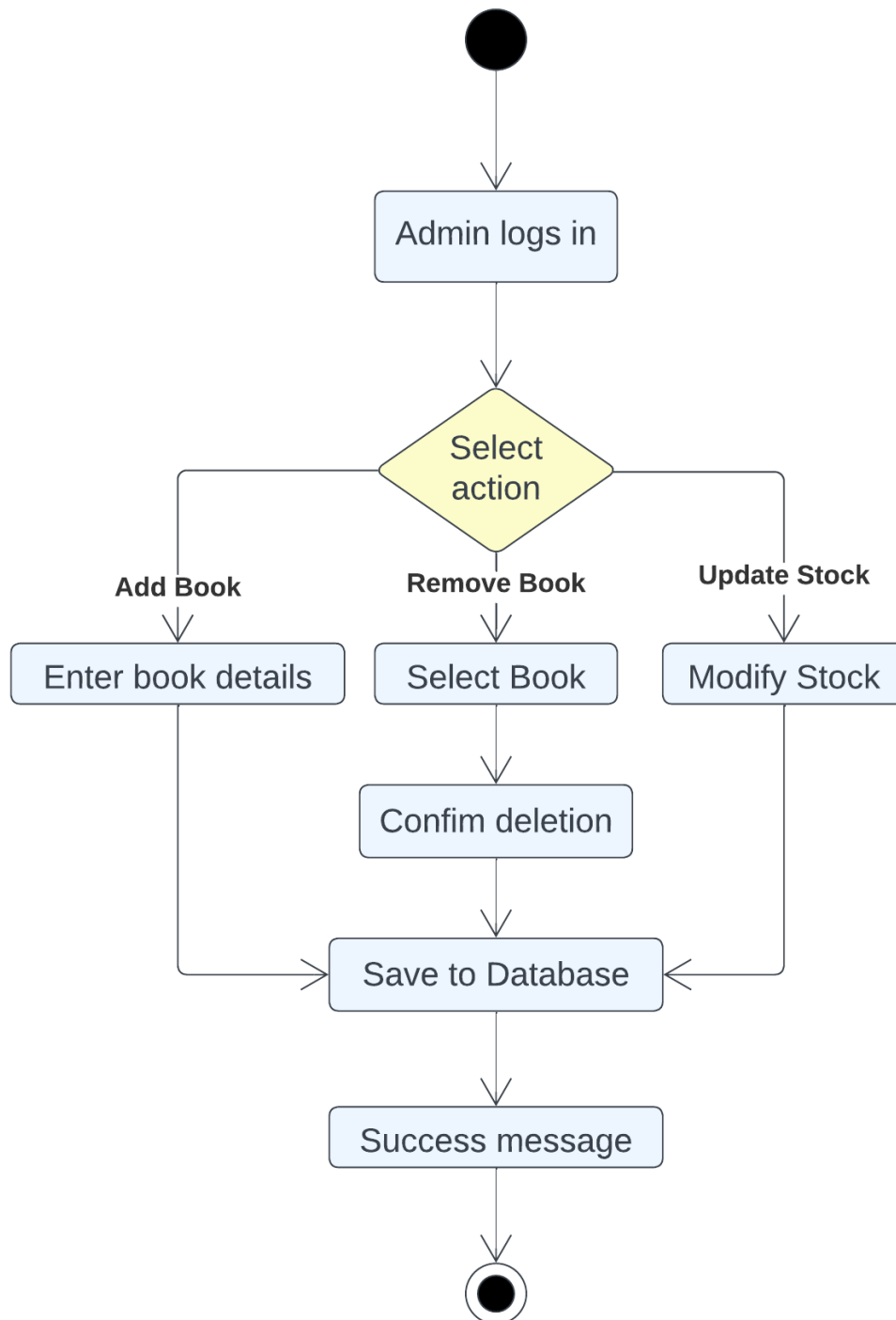


Figure 8: Admin's Book Management System

4.2.4 Order Placement System

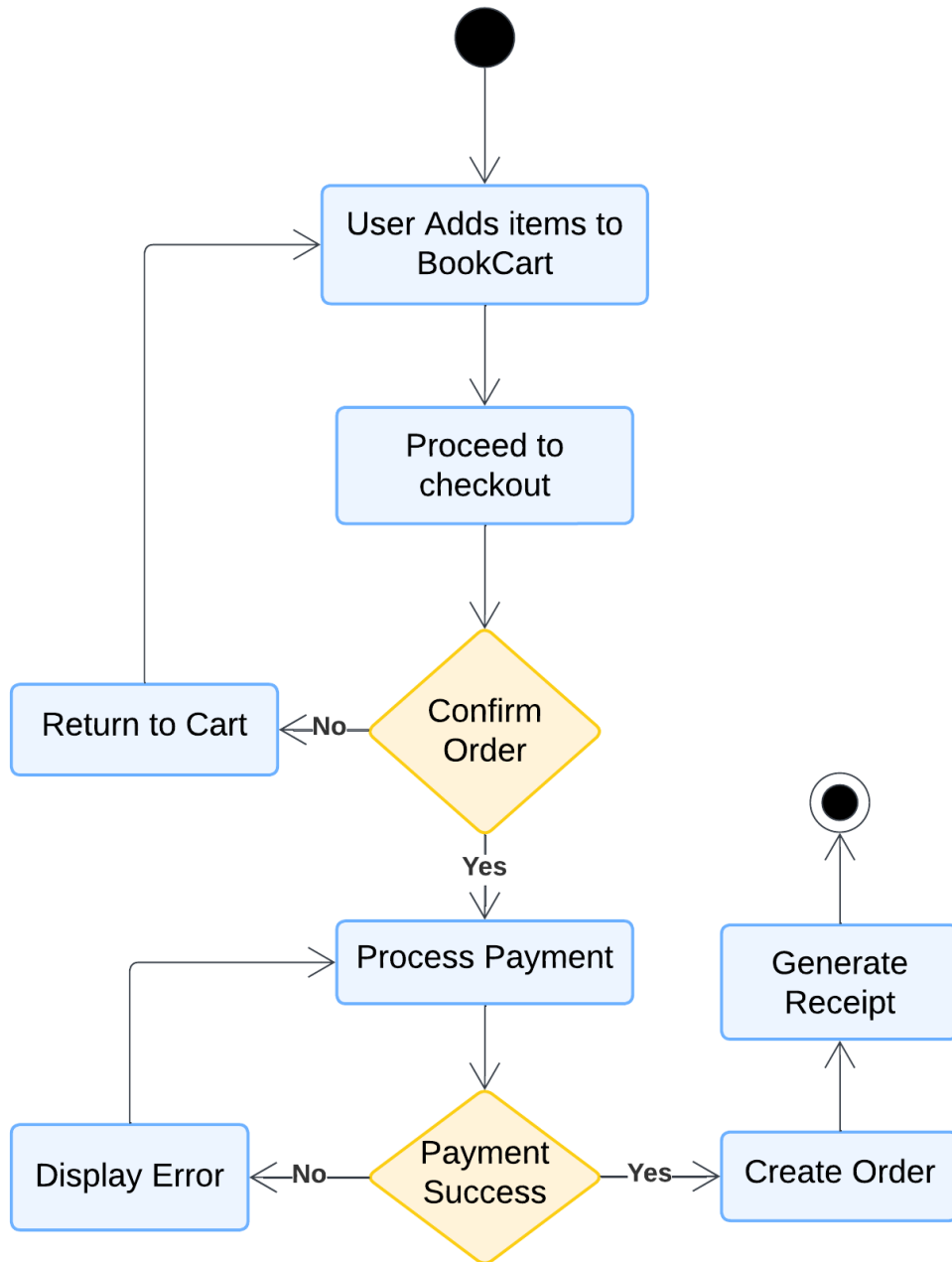


Figure 9: Order placement System

4.2.5 Cart System

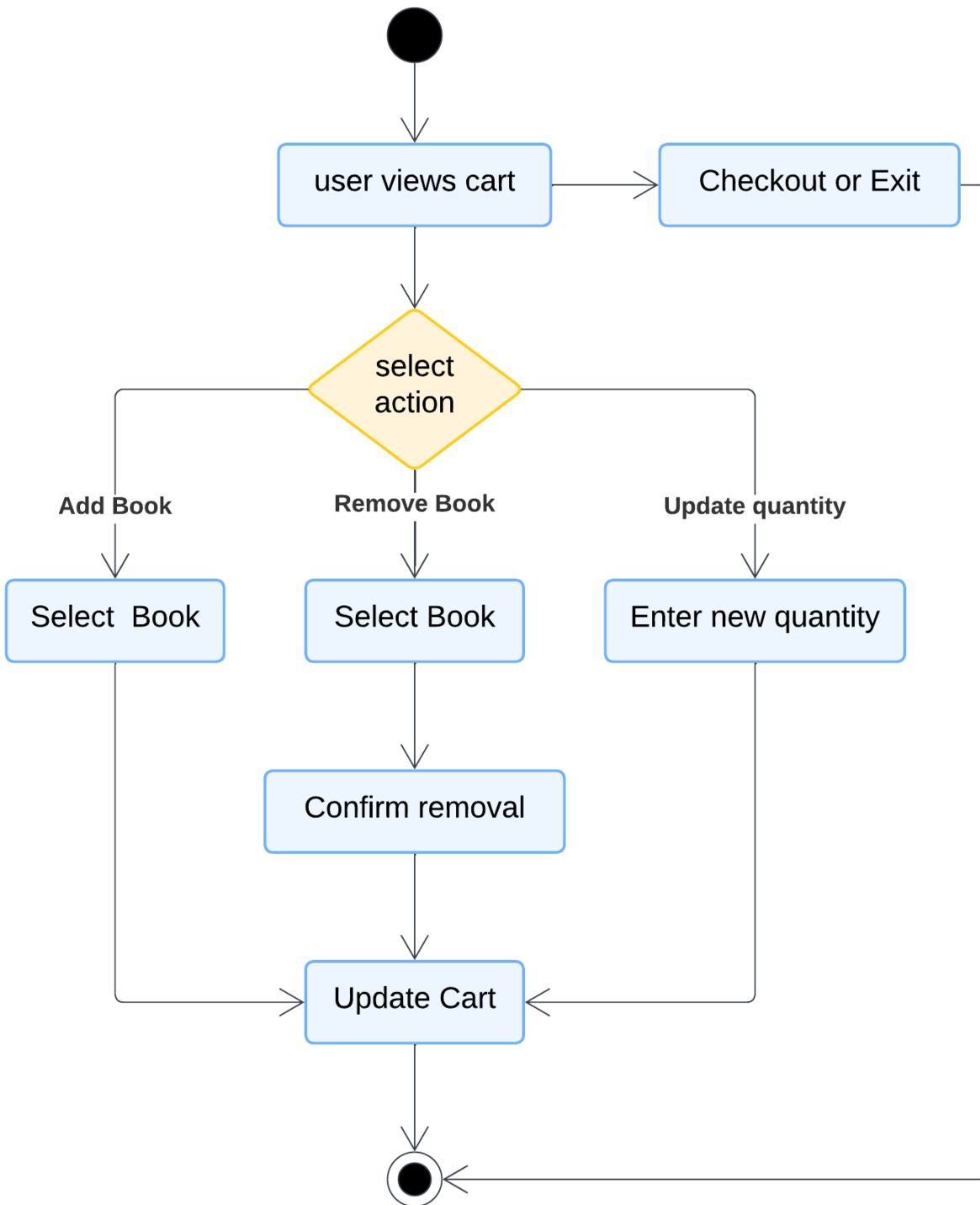


Figure 10: Cart System

4.2.6 Transaction System

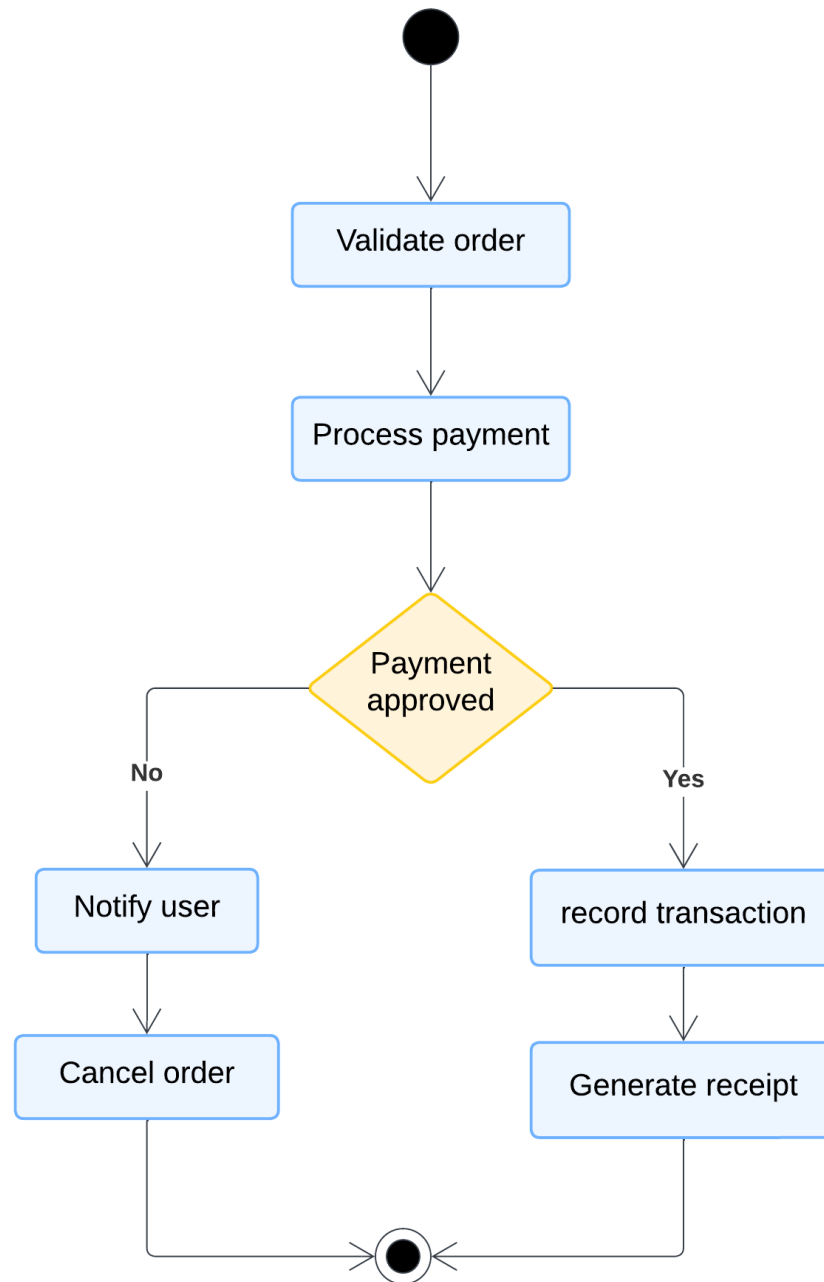


Figure 11: Transaction System

4.3 Sequence Diagram

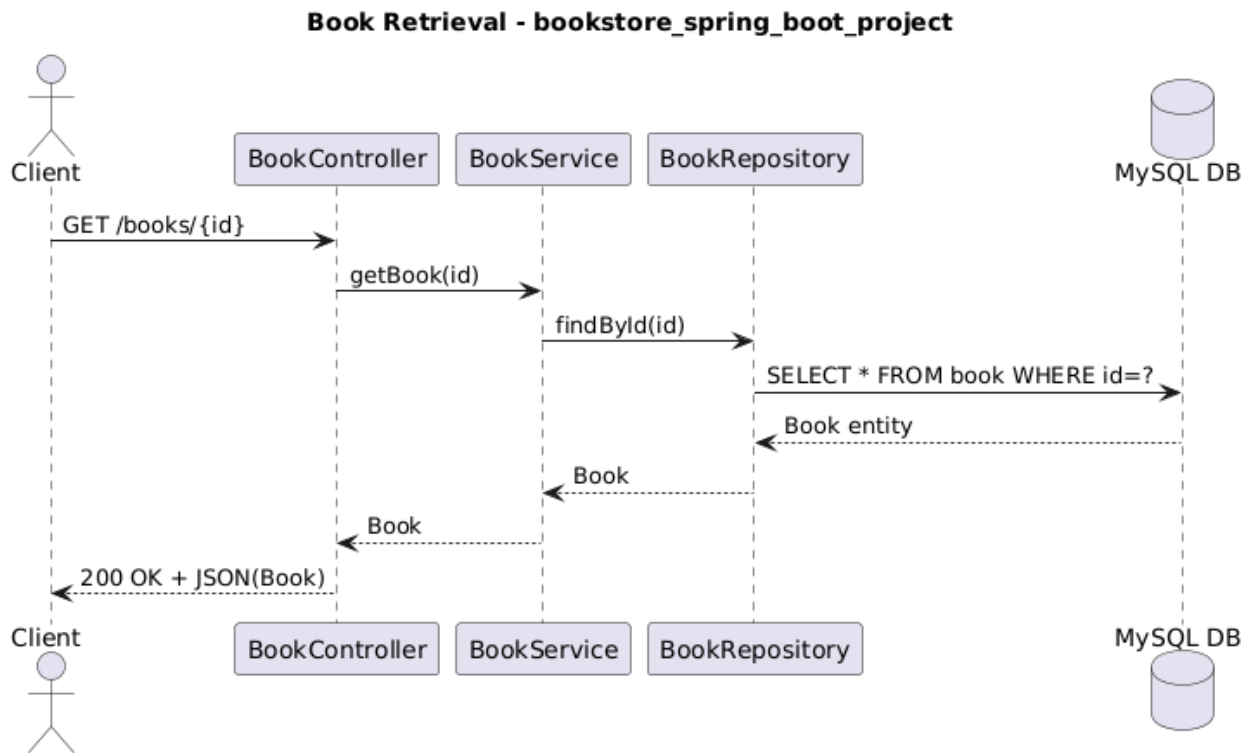


Figure 12: Sequence Diagram

5 User Interface Design

5.1 User Analysis

For Users:

- Users can be anyone accessing the website from computers.
- Average users can read and understand the interface instructions and navigation prompts.
- Users are capable of following written guidance for registration, purchases, and account management.
- The users' age range may span between 10-70 years old.
- No single gender will dominate the user base.
- They use the application when searching for books, making purchases.
- The software is not essential for daily work; most users will access it only occasionally for book purchases.
- The primary interface language is English.
- If users make input mistakes (wrong credentials, invalid payment info), the system will display errors and prompt for correction before proceeding.
- Users are not expected to be experts in inventory systems or e-commerce platforms.
- Some users might be curious about the technology behind the interface, but this knowledge isn't required to use the bookstore features.

For administrators:

- Admin users typically access the system weekly or daily for inventory management.
- They require basic computer literacy but not advanced technical skills.
- The system prevents critical errors through confirmation dialogues (like "Confirm deletion").
- Admin operations focus on book database management rather than complex technical configurations.

5.2 Task Analysis

5.2.1 Task: User Registration

Subtasks:

1. Access registration form
2. Fill in personal details
 - 2.1 Enter valid email
 - 2.2 Create password
 - 2.3 Provide contact information
3. Submit registration

5.2.2 Task: User Login

Subtasks:

1. Access login page
2. Enter credentials
 - 2.1 Input username/email
 - 2.2 Enter password
3. Submit login request
4. Gain access to user account

5.2.3 Task: Admin - Manage inventory

Subtasks:

1. Log in to admin panel
2. Navigate to book management
3. Select managing option
4. Enter book details
 - 4.1 Input title and author
5. Save to database
6. Receive success notification

5.2.4 Task: Place a Book Order

Subtasks:

1. Add books to cart
 - 1.1 Select desired book

- 1.2 Choose quantity
 - 1.3 Click "Add to Cart"
- 2. Proceed to checkout
 - 2.1 View cart contents
 - 2.2 Verify selected items
 - 2.3 Click "Checkout" button
- 3. Confirm order details
 - 3.1 Review shipping information
 - 3.2 Select payment method
 - 3.3 Click "Confirm Order"
- 4. Complete payment
 - 4.1 Enter payment details
 - 4.2 Submit payment
 - 4.3 Receive payment confirmation
- 5. Receive order confirmation
 - 5.1 View order success message

5.2.5 Task: Manage Shopping Cart

Subtasks:

- 1. View cart contents
- 2. Modify items
 - 2.1 Remove unwanted books
 - 2.2 Update quantities
- 3. Save changes
- 4. Continue shopping or proceed to checkout

5.3 Demo Interfaces



Browse books



Log In

Sign Up

Welcome to BookHaven

Meet your next
New Favorite

Browse Books →

About us

Featured Books

Handpicked selections from our curators, featuring the most compelling reads of the season.

Figure 13: Home Page



Browse books



Log In

Sign Up

Create Your Account

Log In

Email

Password

[Forgot Password?](#)

Log In

Sign Up

Figure 14: LogIn Page



Browse books



Log In

Sign Up

Create Your Account

Sign Up

Name

Email

Password

Confirm Password

Create Account

Log In

Figure 15: SignUp Page

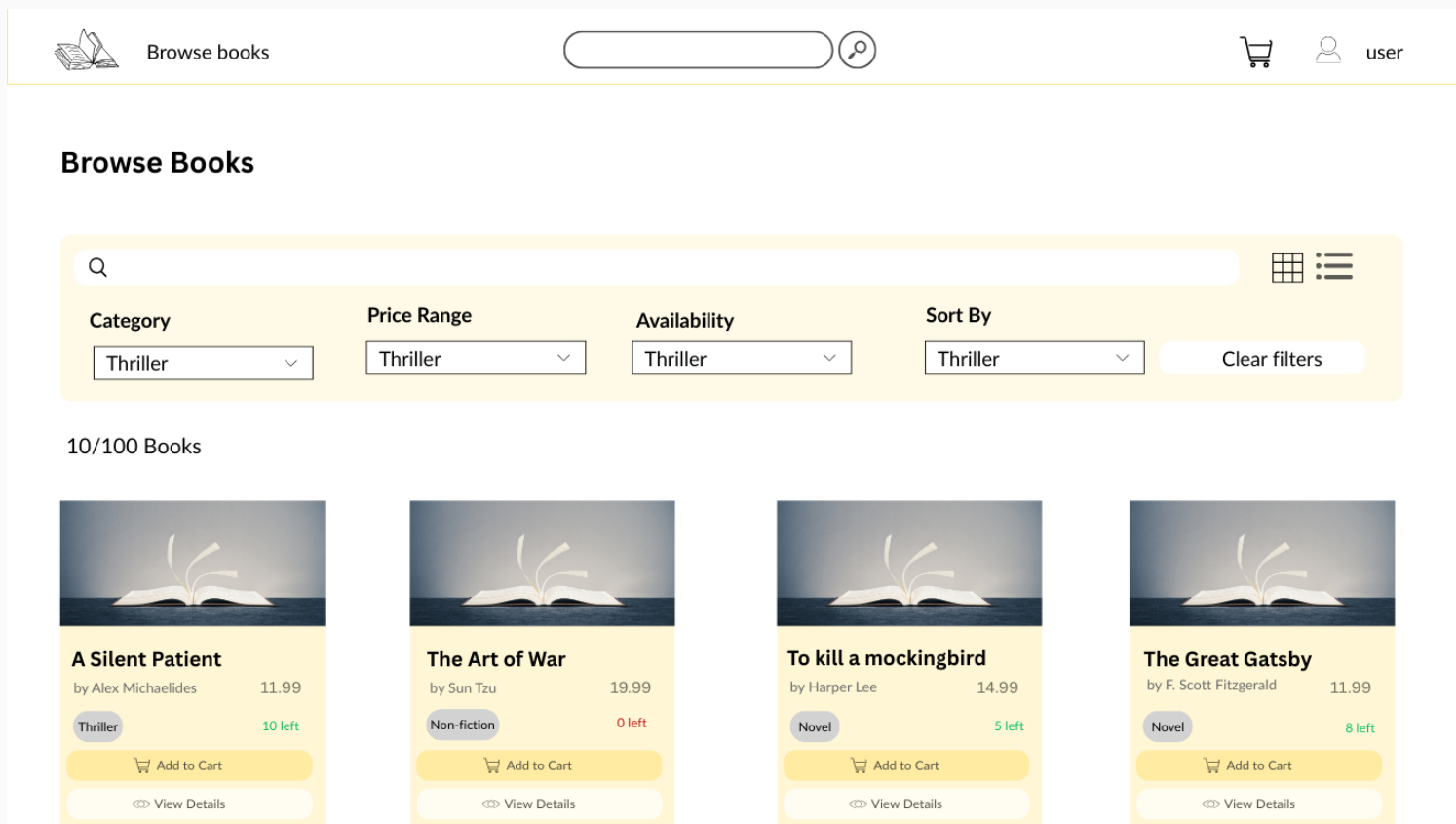


Figure 16: Browse book Page



Browse books



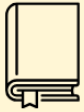
user

← Continue Shopping

Shopping Cart

1 item in your cart

Cart Items



A Silent Patient

by Alex Michaelides

- 1 +

11.99



Order Summary

Subtotal 11.99

Delivery 2.99

Total 14.98

Proceed to Checkout

Figure 17: Shopping Cart Page



Browse books



user

Checkout

1 Shipping — 2 Payment — 3 Review



Shipping information

First Name*

Last Name*

Email*

Phone*

Address*

Continue to payment

Order Summary

Subtotal 11.99

Delivery 2.99

Total 14.98



A Silent Patient

by Alex Michaelides

Qty: 1

Figure 18: Checkout Page



Browse books



user

Checkout

1

Shipping

2

Payment



Payment information

Card Number*

2100 0112 0114 0158

Expiry Date*

MM/YY

CVV*

Name on Card*

☐ Save Card Information

[Back to Shipping](#)

Order Summary

Subtotal 11.99

Delivery 2.99

Total 14.98



A Silent Patient

by Alex Michaelides

Qty: 1

Figure 19: Payment Page

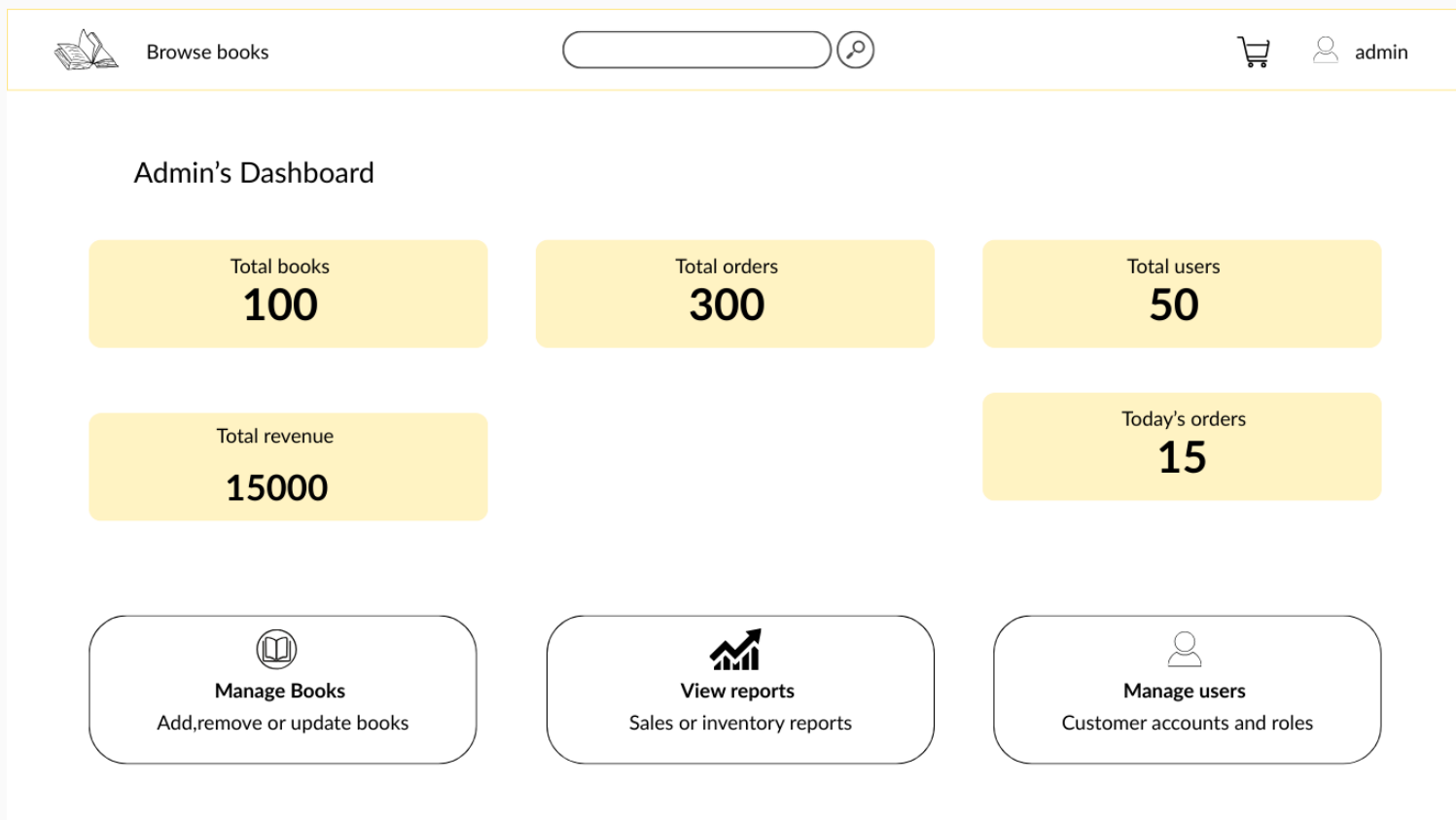


Figure 20: Admin Dashboard Page

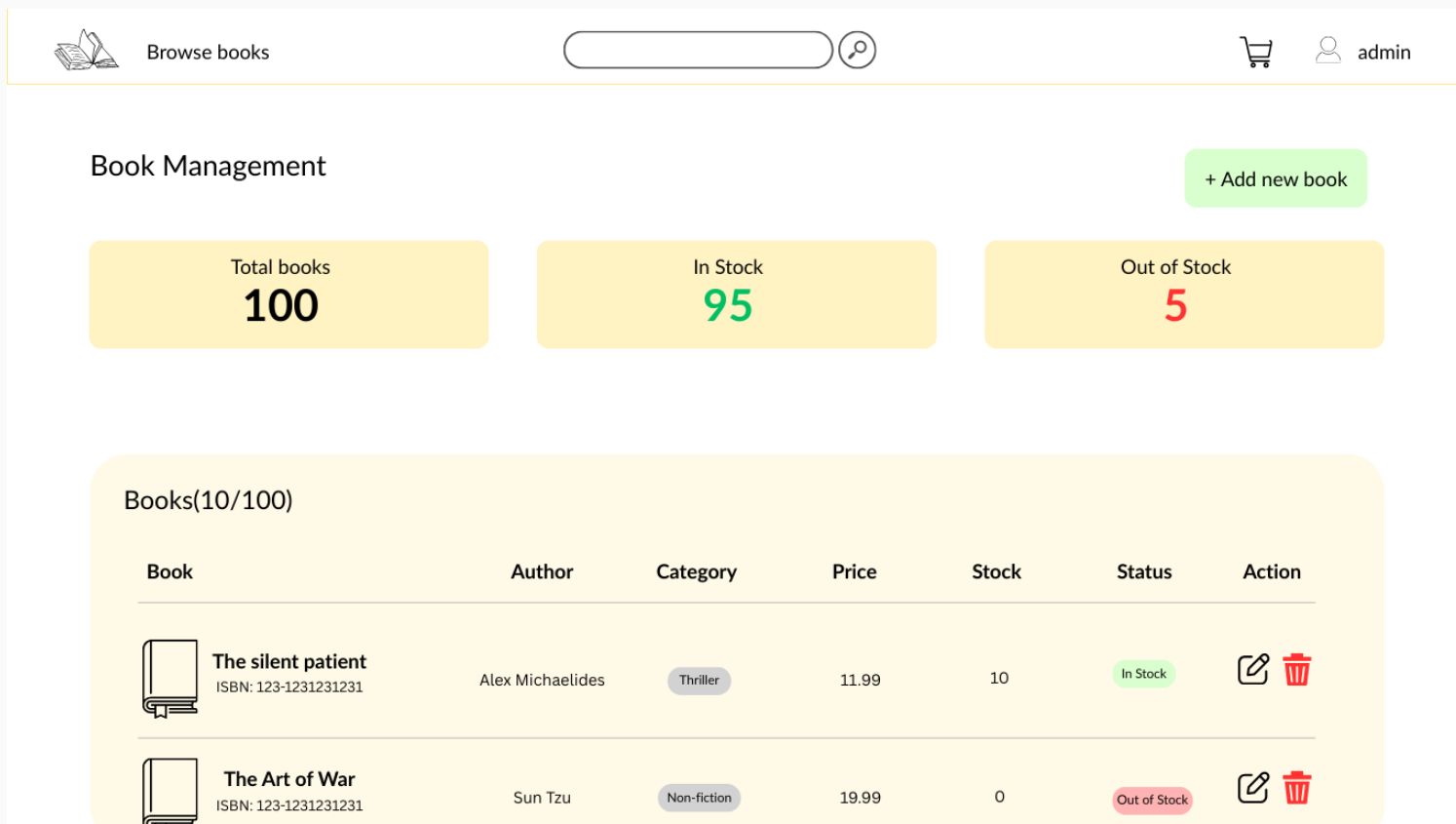


Figure 21: Admin's book managing Page